CCC-C-460a
February 21, 1967
SUPERSEDING
Fed. Spec. CCC-C-460
September 1, 1960

FEDERAL SPECIFICATION

CLOTH, FLANNEL, COTTON (HEAVY, FOR TABLE FELTS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers the requirements for one type of cotton flannel cloth.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

CCC-T-191—Textile Test Methods.
PPP-P-51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

Federal Standard:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington, D. C., 20580.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the finished cloth shall match the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Preproduction sample approval. When required, before production is commenced, a two yard full width sample of the finished cloth shall be submitted or made ready for

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the contracting officer or his authorized representative for inspection as specified. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the production item.

3.3 Material.

3.3.1 *Yarn.* The yarn shall be evenly spun from clean, carded, and drawn cotton, into single yarn for both the warp and filling (see 4.4).

3.3.2 *Physical requirements.* The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

TABLE I. *Physical requirements*

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength grab method (minimum)	
	Warp	Filling	Warp	Filling
Ounces			*Pounds*	*Pounds*
12.5	66	42	60	30

3.3.3 *Width.* The minimum width shall be as specified (see 6.2).

3.3.4 *Weave.* The cloth shall have a filling reversible (or filling back) weave, i.e., one warp and two filling systems, weaving as a 3 down, 1 up regular or broken twill when viewed on either the face or the back (see 4.4).

3.3.5 *Finish.* The cloth shall be unbleached (natural) and shall have a full nap on both sides.

3.3.6 *Length.* Unless otherwise specified (see 6.2), the cloth shall be in one continuous piece of not less than 20 yards for 48 and 54 inch width and not less than 30 continuous yards for 64 and 72 inch width as specified in PPP-P-51.

3.3.7 *Fiber identification.* Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.4 Workmanship. The finished cloth shall conform to the quality and grade of the product established by this specification. Occurence of defects shall not exceed the point level specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of MIL-STD-105 and 4.2.1 through 4.2.3.

4.2.1 *Yard-by-yard examination.* The required yardage of each piece shall be examined on both sides and all defects as defined in section I of Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1. None of the 20 or 30 (whichever is applicable) contiguous linear yards, folded yards, or increments of 1 yard on the measuring device of the inspection machine, selected for examination from any one piece within the sample, shall be penalized more than four points. The sample unit for this examination shall be one linear yard. The sample size shall be 750 yards, regardless of lot size. The sample yardage shall be examined from 25 pieces taken from 25 containers. An approximately equal number of yards shall be examined in each piece of the sample. The lot shall be unacceptable if the points per hundred square yards exceed 32.0. Point computation shall be as follows:

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$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of the material in inches}} = \text{Points per 100 square yards}$$

4.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects up to 3 inches, in any dimension —one point.
For defects exceeding 3 inches, but not exceeding 6 inches in dimension —two points.
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension —three points.
For defects exceeding 9 inches in any dimension —four points.

The following defects, when present, shall be scored four points for each yard in which they occur:

Overall uncleanness.
Objectionable odor.
Degree of napping not equal to, or greater than, the standard sample. Or when no standard sample is employed, the base weave is not obscured by the napped finish.
Width less than specified.

4.2.2 *Examination for length.*

4.2.2.1 *Individual pieces.* During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

4.2.3 *Examination for compliance with Textile Fiber Products Identification Act.* During the yard-by-yard examination, each piece in the sample shall be examined for compliance to the Textile Fiber Products Identification Act. Each piece not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.3 Examination of preparation for delivery. An examination shall be made in accordance with the provisions of PPP-P-51, to determine that the packaging, packing, and marking requirements of section 5 are complied with, except that the inspection level shall be S-2 of MIL-STD-105, and the AQL shall be 2.5 defects per hundred units.

4.4 Testing of the end item. The methods of testing specified in CCC-T-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3, except where otherwise specified, apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be one yard full width of the finished cloth. The sample size (number of sample units) shall be as follows:

Lot size (yards)	Sample size (units)
800 or less	2
801 to 22,000	3
22,001 and over	5

TABLE II. *Instructions for testing*

Characteristic	Requirement reference	Test method
Fiber identification[1]	3.3.1	1200
Weave	3.3.4	Visual[2]
Weight	Table I	5041
Yarns per inch:		
Warp	Table I	5050
Filling	Table I	5050
Breaking strength:		
Warp	Table I	5100
Filling	Table I	5100

[1] Supplier's certificate of compliance will be accepted for this characteristic.
[2] One determination shall be made on each sample unit and the result reported as "pass" or "fail".

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C (see 6.2 and 6.2.1).

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5.1.1 *Levels A and C.* The cloth, put up as specified (see 3.4), shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-51.

5.2 **Packing.** Packing shall be level A, B, or C (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP-P-51.

5.3 **Marking.** In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the applicable requirements of PPP-P-51.

6. NOTES

6.1 **Intended use.** The cloth covered by this specification is intended for use as table felts.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Width of cloth required (see 3.3.3).
(*c*) Length of rolls if other than specified (see 3.6).
(*d*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in PPP-P-51, may be exercised in the preparation for the delivery of cloth.

6.3 **Standard sample.** For access to standard sample, address the procuring officer issuing the invitation for bids.

6.4 When level B packaging requirements are specified by civil agency procurement officers, the requirements of 5.1.1 shall apply.

CUSTODIAN:
Navy—SA
Air Force—67

INTERESTED ACTIVITIES:

Review:
Air Force—67

User:
Navy—MC, CG, BUMED

Preparing activity:
Navy—SA

Review and user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
CCC-C-460a Cloth, Flannel, Cotton (Heavy For Table Felts)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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